

New sampling formulae related to linear canonical transform[☆]

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Abstract

Linear canonical transform (LCT) is an integral transform with four parameters a, b, c, d and has been shown to be a powerful tool for optics, radar system analysis, filter design, phase retrieval, pattern recognition, and many other applications. Many well-known transforms such as the Fourier transform, the fractional Fourier transform, and the Fresnel transform can be seen as special cases of the linear canonical transform. In this paper, new sampling formulae for reconstructing signals that are band-limited or time-limited in the linear canonical transform sense have been proposed. Firstly, the sampling theorem representation of band-limited signals associated with linear canonical transform from the samples taken at Nyquist rate is derived in a simple way. Then, based on the relationship between the Fourier transform and the linear canonical transform, the other two new sampling formulae using samples taken at half the Nyquist rate from the signal and its first derivative or its generalized Hilbert transform are obtained. The well-known sampling theorems in Fourier domain or fractional Fourier domain are shown to be special cases of the achieved results. The experimental results are also proposed to verify the accuracy of the obtained results. Finally, discussions about these new results and future works related to the linear canonical transform are proposed.

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1. Introduction

The linear canonical transform (LCT) [1–5] is an integral transform with four parameters a, b, c, d . It was first introduced in 1970s and has found many applications in optics, radar system analysis, filter design, phase retrieval, pattern recognition, and

many other applications [3–5]. Many transforms such as the Fourier transform (FT), the fractional Fourier transform (FrFT), Fresnel transform, and scaling operations are all special cases of the LCT [4–7]. In some papers, the LCT is also known as the affine Fourier transform (AFT) [8], the generalized Fresnel transform [9] and the ABCD transform [2].

Simultaneously, sampling process is the basis theory for converting the continuous signals to discrete signals, and it is central in almost any domain because it provides the link between the continuous physical signals and the discrete domain. The sampling theorem expansions for the linear canonical transform of compact functions in time domain or linear canonical transform domain have been derived recently by Stern

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[10] and Deng [11], respectively. The formulae derived in [10] and [11] can be used to reconstruct the original signal or its linear canonical transform from their samples at a discrete set of points that satisfying the Nyquist rate of that particular domain.

However, the sampling theorem derived in [10] is not in an interpolation form strictly and the derivation of the sampling theorem in [11] is somewhat complex. Moreover, in some real applications, when sampling broad-band or non-stationary signals, the sampling rate may not satisfy the condition of sampling theorem of that particular domain [10,11]. To overcome the above shortcomings, it is interesting to explore new sampling formulae that keep the sampling rate as low as possible [12]. This can be achieved, for example, by taking a sampling series involving sampled values of the signal and its first derivative. In this case, it suffices to take samples below the Nyquist rate of that domain. The potential applications involving this arises in radar and sonar signal processing or digital flight control where the velocity as well as position is recorded. Therefore, it is worthwhile and interesting to obtain new sampling formulae associated with the linear canonical transform that keeps the sampling rate as low as possible.

In this paper, a simple representation of sampling theorem proposed in [10,11] is first derived based on a useful relationship between the linear canonical transform and the Fourier transform. Then two new sampling formulae related to the linear canonical transform are derived. The new sampling formulae can be used to reconstruct the signal or its linear canonical transform from the samples of the signal and its first derivative or its generalized Hilbert transform, each at half of the Nyquist rate. Furthermore, the classical results achieved in [13–16] of Fourier domain or fractional Fourier domain [17–19] can be seen as special cases of our achieved results.

2. Preliminary

2.1. The LCT

The LCT with parameter $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ of a signal $f(t)$ is defined as [1,3]

$$\begin{aligned} L_f^{\mathbf{A}}(u) &= L_f^{\mathbf{A}}[f(t)](u) \\ &= \begin{cases} \sqrt{\frac{1}{j2\pi b}} \int_{-\infty}^{+\infty} f(t) e^{j\frac{1}{2}[\frac{a}{b}t^2 - (\frac{c}{b}t + \frac{d}{b}u^2)]} dt, & b \neq 0, \\ \sqrt{d} e^{j\frac{1}{2}cd u^2} f(du), & b = 0. \end{cases} \end{aligned} \quad (1)$$

The inverse transform for LCT is given by a linear canonical transform having parameters

$$\mathbf{A}^{-1} = \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \text{ that is}$$

$$f(t) = \begin{cases} \sqrt{\frac{1}{-j2\pi b}} \int_{-\infty}^{+\infty} L_f^{\mathbf{A}}(u) e^{j\frac{1}{2}[\frac{a}{b}u^2 + (\frac{c}{b}tu - \frac{d}{b}t^2)]} du, & b \neq 0, \\ \sqrt{a} e^{-j\frac{1}{2}cat^2} f(at), & b = 0. \end{cases} \quad (2)$$

It is noted that when $b = 0$, the LCT of a signal is essentially a chirp multiplication and it is of no particular interest for our objective in this work, so it will not be discussed in this paper. And without loss of generality, we assume $b > 0$ in the preceding sections. For further details about the definition and properties of LCT, [1–5] can be referred.

A signal $f(t)$ is band-limited to $\Omega_{\mathbf{A}}$ in a linear canonical transform sense, which means that:

$$L_f^{\mathbf{A}}(u) = 0 \quad \text{for } |u| > \Omega_{\mathbf{A}}, \quad (3)$$

where $\Omega_{\mathbf{A}}$ is called the bandwidth of the signal $f(t)$ in the linear canonical transform domain.

2.2. The generalized Hilbert transform

Hilbert transform is also found to play an important role in signal processing and optical analysis. The Hilbert transform of a signal $f(t)$ is defined as

$$H\{f(t)\}(y) = \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{f(t)}{y - t} dt. \quad (4)$$

The analytic part of $f(t)$ is defined through the Hilbert transform of the signal as

$$F(t) = f(t) + jH[f](t). \quad (5)$$

One of the most important properties for analytical signals is that they contain no negative Fourier frequency components. The expansion of Hilbert transform and associated analytical signal for fractional Fourier transform domain that has similar properties with the Fourier transform domain has been proposed by Zayed in [20].

Along this way, the author generalized the definition of Hilbert transform of linear canonical transform domain [21] as

$$H_{\mathbf{A}}[f](t) = \frac{e^{-j(d/2b)t^2}}{\pi} \int_{-\infty}^{+\infty} \frac{f(x) e^{j(a/2b)x^2}}{t - x} dx. \quad (6)$$

And the definition of analytical signal in linear canonical transform domain as

$$\tilde{F}_A(t) = f(t) + jH_A[f](t).$$

The above generalized definitions of Hilbert transform and analytical signal in [21] has similar properties in linear canonical transform domain with respect to the Hilbert transform in Fourier transform domain. That is to say they contain no negative components in the particular linear canonical transform domain.

3. Main results

Before we introduce the main results of the paper, a useful lemma associated with a band-limited signal $f(t)$ of linear canonical transform domain is derived firstly.

Lemma 1. Assume a signal $f(t)$ is band-limited to Ω_A in linear canonical transform domain with parameter $A = \{a, b, c, d\}$ and $b > 0$. Let

$$g(t) = \int_{-\Omega_A}^{\Omega_A} L_f^A(u) e^{-j(d/2b)u^2 + j(1/b)ut} du. \quad (7)$$

Then, $g(t)$ is a signal band-limited to $(1/b)\Omega_A$ in conventional Fourier domain.

Proof. Applying the Fourier transform for both sides of Eq. (7), it is easy to obtain:

$$\begin{aligned} F(g(t))(w) &= \int_{-\infty}^{\infty} \int_{-\Omega_A}^{\Omega_A} L_f^A(u) e^{-j(a/2b)u^2 + jut(1/b)} du e^{-jw t} dt \\ &= \int_{-\Omega_A}^{\Omega_A} L_f^A(u) e^{-j(a/2b)u^2} \int_{-\infty}^{\infty} e^{-jw t + jut(1/b)} dt du \\ &= \int_{-\Omega_A}^{\Omega_A} L_f^A(u) e^{-j(a/2b)u^2} \frac{1}{2\pi} \delta\left(-w + \frac{1}{b}u\right) du \\ &= \frac{1}{2\pi} L_f^A(bw) e^{-j(ab/2)w^2} \end{aligned}$$

Because $L_f^A(u)$ is Ω_A band-limited, so $|bw| < \Omega_A$, that is $|w| < \Omega_A/b$. Then, from the definition of band-limited signal in traditional Fourier domain, $g(t)$ is a signal band-limited to $(1/b)\Omega_A$ in Fourier domain. $\square$

3.1. Sampling theorem of band-limited signal in LCT domain

Recently, Stern [10] and Deng [11] proposed the sampling theorem expansion related to the linear

canonical transform. The following formula was proposed in [11]:

Theorem 1. Let signal $f(t)$ is band-limited to Ω_A in linear canonical transform domain having parameter $A = \{a, b, c, d\}$ and $b > 0$. Then, the following sampling theorem expansion for $f(t)$ holds:

$$f(t) = e^{-j(a/2b)t^2} \sum_{n=-\infty}^{+\infty} f(nT) e^{j(a/2b)(nT)^2} \times \text{sinc}\left[\frac{\Omega_A(t - nT)}{b\pi}\right], \quad (8)$$

where T is the sampling period and satisfies

$$T = \pi b / \Omega_A$$

and $f = \Omega_A/\pi b$ is called as the Nyquist rate of sampling theorem associated with the linear canonical transform.

Proof. A more elegant proof of this theorem than that of [10,11] is now given as follows. The idea is to use the relationship between the linear canonical transform and the Fourier transform that was reflected in Lemma 1 above.

From Lemma 1, the original signal can be represented as following:

$$f(t) = \sqrt{\frac{1}{-j2\pi b}} e^{-j(a/2b)t^2} g(t), \quad (9)$$

where $g(t)$ is band-limited to $(1/b)\Omega_A$ in conventional Fourier domain, therefore, the classical Shannon's sampling theorem can be applied to $g(t)$, that is

$$g(t) = \sum_{n=-\infty}^{+\infty} g(nT) \text{sinc}\left[\frac{\Omega_A(t - nT)}{b\pi}\right], \quad (10)$$

where $T = \pi b / \Omega_A$ is the sampling period. By putting Eq. (10) in Eq. (9), the final result is thus obtained. $\square$

Eq. (8) can be looked as the sampling theorem of band-limited signals associated with the linear canonical transform. By exchanging the role of the signal and its LCT, the sampling theorem of time-limited signal in LCT domain derived in [10] can be easily obtained from Theorem 1. It should be noted that the results derived in Eq. (8) is more explicit than that in [10], and the derivation is easier than that in [11]. Furthermore, when the signal $f(t)$ is band-passed to $[w_0 - \Omega_A, w_0 + \Omega_A]$ in the linear canonical transform sense, the following corollary is obtained.

Corollary 1. If a signal $f(t)$ band passes to $[w_0 - \Omega_A, w_0 + \Omega_A]$ in linear canonical transform domain with parameter $\mathbf{A} = \{a, b, c, d\}$ and $b > 0$. Then the following sampling theorem expansion for $f(t)$ holds as

$$f(t) = e^{-j\frac{a}{2b}t^2} \sum_{n=-\infty}^{+\infty} f(nT) e^{j(a/2b)(nT)^2} \text{sinc} \left[\frac{\Omega_A(t - nT)}{b\pi} \right] \times e^{j(w_0/b)(t - nT)}, \quad (11)$$

where $T = \pi b / \Omega_A$ is the sampling period associated with the linear canonical transform.

Eqs. (8) and (11) are the sampling theorem expansion of the classical results in linear canonical transform domain, and when the LCT parameter reduces to $\mathbf{A} = \{\cos \theta, \sin \theta, -\sin \theta, \cos \theta\}$, the result in Eq. (8) reduces to the sampling theorem of fractional Fourier domain [17–19], which in a specific case $\theta = \pi/2$, becomes the classical sampling theorem of Fourier domain. The results in Eq. (11) can also be looked as the generalization the results in [17] of fractional Fourier domain.

Though the sampling theorem introduced above works well in practice, in some real applications, when sampling non-stationary signals or broadband signals, the sampling condition in Eq. (8) may not always hold true, therefore, it is interesting and worthwhile to derive new sampling formulae and keeping the sampling rate as low as possible. In the following sections the other two new sampling formulae that involving the signal and its first derivate or its generalized Hilbert transform associated with the LCT are derived.

3.2. Sampling formula involving the signal and its derivate

If a signal $f(t)$ has continuous first derivative, then the following sampling theorem associated with the linear canonical transform can be derived.

Theorem 2. Assume a signal $f(t)$ and its first derivative are continuous, $f(t)$ is band-limited to Ω_A in linear canonical transform sense. Then the following sampling formula for $f(t)$ holds

$$f(t) = e^{-j(a/2b)t^2} \sum_{k=-\infty}^{\infty} e^{j(a/2b)(kT_1)^2} \times \left\{ \left[1 + j(kT_1)(t - kT_1) \frac{a}{b} \right] \right.$$

$$\left. \times f(kT_1) + (t - kT_1)f'(kT_1) \right\} \times \left[\text{sinc} \left(\frac{t}{T_1} - k \right) \right]^2, \quad (12)$$

where $T_1 = 2T = 2\pi b / \Omega_A$, $f'(kT_1) = x'(t)|_{t=kT_1}$.

Proof. Since the signal $f(t)$ is band-limited to Ω_A in fractional Fourier sense. Then from Lemma 1 and the definition of inverse LCT, the signal $f(t)$ can be represented as

$$f(t) = \sqrt{\frac{1}{-j2\pi b}} e^{-j(a/2b)t^2} g(t), \quad (13)$$

where $g(t)$ is a signal band-limited to $\Omega = (1/b)\Omega_A$ in the conventional Fourier domain, therefore the classic sampling results in [12,13] can be applied to the signal $g(t)$, and obtaining the following sampling formula:

$$g(t) = \sum_{k=-\infty}^{\infty} \left\{ g\left(\frac{2k\pi}{\Omega}\right) + \left(t - \frac{2k\pi}{\Omega}\right) g'\left(\frac{2k\pi}{\Omega}\right) \right\} \times \left[\text{sinc} \frac{1}{2} \left(\frac{\Omega t}{\pi} - 2k \right) \right]^2$$

and

$$g(t) = \sqrt{-j2\pi b} e^{j(a/2b)t^2} f(t),$$

$$g'(t) = \sqrt{-j2\pi b} e^{j(a/2b)t^2} \left[jtf(t) \frac{a}{b} + f'(t) \right].$$

Substituting the above equations in Eq. (13) and obtaining the final results. $\square$

Corollary 2. If the parameter of LCT reduces to $\mathbf{A} = \{\cos \theta, \sin \theta, -\sin \theta, \cos \theta\}$, then the sampling theorem related to fractional Fourier transform domain can be obtained as

$$f(t) = e^{-j(1/2)t^2 \cot \alpha} \sum_{k=-\infty}^{\infty} e^{j(1/2)(kT_1)^2 \cot \alpha} \times \{ [1 + jkT_1(t - kT_1) \cot \alpha] f(kT_1) + (t - kT_1)f'(kT_1) \} \left[\text{sinc} \left(\frac{t}{T_1} - k \right) \right]^2. \quad (14)$$

Eqs. (12)–(14) provide alternative sampling formulae for reconstruction of signals that are band-limited in LCT domain or fractional Fourier transform domain from the samples of the signal and its first derivative, each at half of the Nyquist rate. The potential applications of this sampling theorem occur in radar and sonar signal processing as well as digital flight control where the velocity as

well as position is recorded, and the velocity can be seen as the first derivative of position. Moreover, the classical results in [14,15] by Fogel can also be looked as special cases of Theorem 2.

3.3. Sampling formula involving the signal and its GHT

The following sampling formula associated with the signal $f(t)$ and its generalized Hilbert transform (GHT) $H_A[f](t)$ [21] can be looked as an extension of the results in [22] associated with the fractional Fourier transform.

Theorem 3. Let a signal $f(t)$ band-limited to Ω_A in linear canonical transform sense. Then $f(t)$ can be represented by the samples of $f(t)$ and its generalized Hilbert transform $H_A[f](t)$ at half Nyquist rate as follows:

$$\begin{aligned} f(t) = & e^{-j(a/2b)t^2} \sum_{k=-\infty}^{+\infty} \left\{ e^{j(a/2b)(kT_2)^2} f(kT_2) \right. \\ & \times \cos \left[\frac{\pi}{T_2}(t - kT_2) \right] - e^{j(d/2b)(kT_2)^2} H_A[f](kT_2) \\ & \left. \times \sin \left[\frac{\pi}{T_2}(t - kT_2) \right] \right\} \operatorname{sinc} \left[\frac{\pi}{T_2}(t - kT_2) \right], \end{aligned} \quad (15)$$

where $T_2 = 2T = 2\pi b/\Omega_A$.

Proof. From Lemma 1, $g(t)$ is band-limited to $\Omega = (1/b)\Omega_A$ in the conventional Fourier domain, applying the well-known formulae associated with the traditional Hilbert transform in Fourier transform domain [12,16], we get

$$\begin{aligned} g(t) = & \sum_{k=-\infty}^{+\infty} \left\{ g\left(\frac{2k\pi}{\Omega}\right) \cos\left(\frac{1}{2}(\Omega t - 2k\pi)\right) \right. \\ & \left. - \tilde{g}\left(\frac{2k\pi}{\Omega}\right) \sin\left(\frac{1}{2}(\Omega t - 2k\pi)\right) \right\} \\ & \times \operatorname{sinc}\left(\frac{1}{2\pi}(\Omega t - 2k\pi)\right), \end{aligned} \quad (16)$$

where $\tilde{g}(t) = (1/\pi) \int_{-\infty}^{+\infty} (g(v)/t - v) dv$ is the traditional Hilbert transform of signal $g(t)$.

From the definition of traditional Hilbert transform we obtain

$$\begin{aligned} \tilde{g}(t) = & \frac{1}{\pi} \int_{-\infty}^{+\infty} \frac{g(v)}{t - v} dv \\ = & \frac{1}{\pi} \sqrt{-j2\pi b} \int_{-\infty}^{+\infty} \frac{f(v) e^{j(a/2b)v^2}}{t - v} dv \\ = & \sqrt{-j2\pi b} e^{j(d/2a)t^2} H_A[f](t). \end{aligned} \quad (17)$$

Substituting it in Eq. (16) and obtaining

$$\begin{aligned} g(t) = & \sqrt{-j2\pi b} \sum_{k=-\infty}^{+\infty} \left\{ e^{j(b/2a)(kT_2)^2} f(kT_2) \right. \\ & \times \cos \left[\left(\frac{\pi}{T_2} t - k\pi \right) \right] - e^{j(d/2a)(kT_2)^2} H_A[f](kT_2) \\ & \left. \times \sin \left[\frac{\pi}{T_2} t - k\pi \right] \right\} \operatorname{sinc} \left[\frac{\pi}{T_2} t - k\pi \right]. \end{aligned} \quad (18)$$

Substituting Eq. (18) into Eq. (13) and obtain the final result. $\square$

Eq. (17) can be seen as a generalization of Goldman's formula as in [16] and it reduces to it when $A = \{0, 1, -1, 0\}$. Simultaneously, we also get the following useful corollary.

Corollary 3. If the parameter of LCT changes to $A = \{\cos \theta, \sin \theta, -\sin \theta, \cos \theta\}$, then the sampling formula of Theorem 3 reduces to the sampling formula in fractional Fourier transform domain as follows:

$$\begin{aligned} f(t) = & e^{-j(\cot \alpha/2)t^2} \sum_{k=-\infty}^{+\infty} \left\{ e^{j(\cot \alpha/2)(kT_2)^2} f(kT_2) \right. \\ & \times \cos \left[\frac{\pi}{T_2}(t - kT_2) \right] - e^{j(\cot \alpha/2)(kT_2)^2} H_\alpha[f](kT_2) \\ & \left. \times \sin \left[\frac{\pi}{T_2}(t - kT_2) \right] \right\} \operatorname{sinc} \left[\frac{\pi}{T_2}(t - kT_2) \right], \end{aligned} \quad (19)$$

where $H_\alpha[f](nT_2)$ denotes the generalized Hilbert transform in fractional Fourier transform domain [20] of signal $f(t)$.

Especially, if a signal $y(t)$ can be written as $y(t) = f(t)e^{j(a/2b)t^2}$, and $f(t)$ satisfies the condition of Theorem 3, then the following result can be deduced:

$$\begin{aligned} y(t) = & \sum_{n=-\infty}^{+\infty} \left\{ y(nT_2) \cos \left[\frac{\Omega_A}{2b}(t - nT_2) \right] \right. \\ & \left. - H[y](nT_2) \sin \left[\frac{\Omega_A}{2b}(t - nT_2) \right] \right\} \\ & \times \operatorname{sinc} \left[\frac{\Omega_A(t - nT_2)}{2\pi b} \right], \end{aligned} \quad (20)$$

where $H[y](nT_2)$ denotes the classical Hilbert transform of signal $y(t)$.

The sampling formula of Eq. (19) is a new formula related to the fractional Fourier transform, and the result in Eq. (20) is also a novel results related to LCT, and when $A = \{\cos \theta, \sin \theta, -\sin \theta, \cos \theta\}$, the results in Eq. (20) reduces to the well known results in [22] of fractional Fourier domain.

4. Discussion

In proving the above theorems, a technique which has been widely investigated by the fractional Fourier transform literature [17–19,23,24] is used, continuing this path we can also extend the classical results in Fourier transform domain [12,16,25,26] to the linear canonical transform domain based on the relationship between the Fourier transform and LCT. For example, if a signal $f(t)$ is Ω_A band-limited in LCT domain with parameter $A = \{a, b, c, d\}$, the values of $f(t)$ and its first $R - 1$ derivatives are known

on the sequence $\{\lambda_n | \lambda_n = (Rn\pi/\Omega_A), n \in Z\}$, then $f(t)$ is completely determined for all $t \in (-\infty, +\infty)$, simultaneously the sampling rate can be set to $(1/R)$ th Nyquist rate in this domain. How to obtain clear representation formulae in this condition and the other sampling formulae in linear canonical transform domain are also our future works.

5. Experimental results

Figs. 1–4 plots the results of numerical experiments demonstrating the effectiveness of the sampling

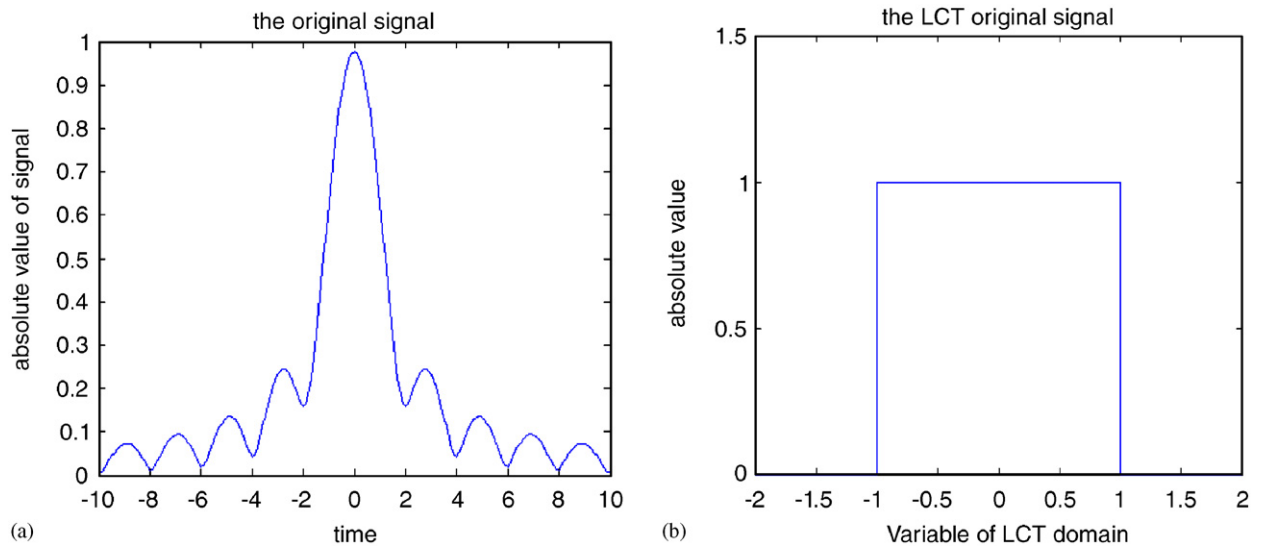


Fig. 1. (a) Absolute value of signal $f(t)$, (b) the LCT of signal $f(t)$ with parameters $A = \{1, 2/\pi, 0, 1\}$.

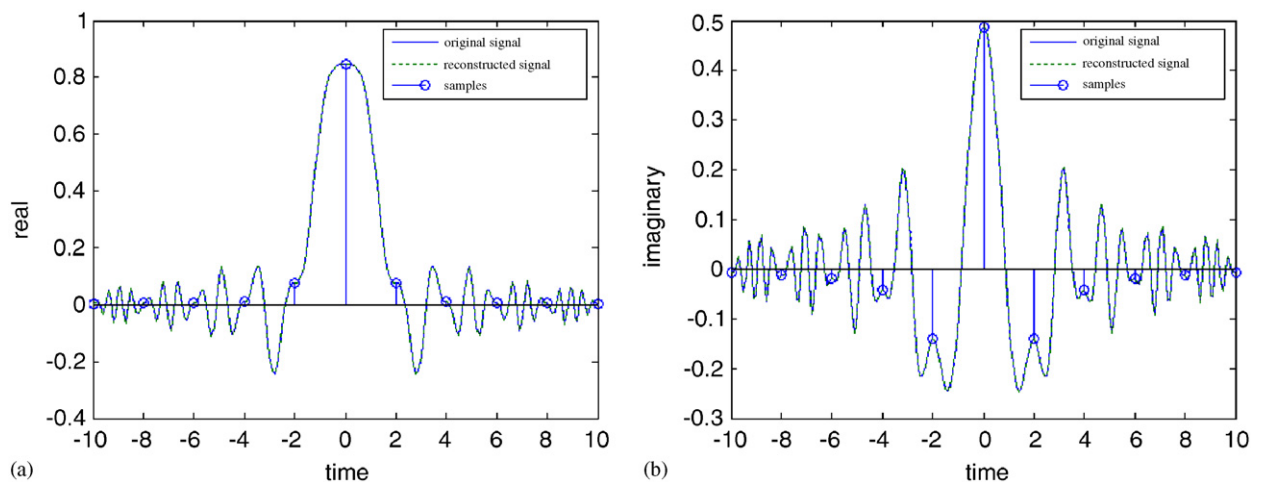


Fig. 2. (a) Real part of $f(t)$, of the samples of $f(t)$ and of the reconstructed signal by the signal samples at the Nyquist rate of LCT domain. (b) Imaginary part of $f(t)$, of the samples of $f(t)$ and of the reconstructed signal the signal samples at the Nyquist rate of LCT domain. It is noted from Fig. 2 that the reconstructed signal denoted by dotted line overlaps exactly the original signal denoted by continuous line.

formulae presented in this paper. Figs. 1(a, b) show a LCT pair with $\mathbf{A} = \{1, 2/\pi, 0, 1\}$ of signal $f(t)$, and the LCT of signal $f(t)$ is a real rectangular function confined to the interval $(-1, 1)$. Therefore, from Theorems 1–3 we can reconstruct the original signal by sampling interval $T = \pi b / \Omega_A = 2$, $T_1 = 2T = 4$ and $T_2 = 2T = 4$, respectively. The real and imaginary parts of the original, sampled and reconstructed signals based on Theorems 1–3 are plotted in Figs. 2(a, b), Figs. 3(a, b) and 4(a, b), respectively. In the

figures, reconstructed signal is denoted by dotted line, and the original signal denoted by the continuous line.

These experiment results demonstrate that we can perfect reconstruct the original signal by the sampling formula derived in Theorem 1. And performance of Theorem 1 is the best among these three sampling theorems, the sampling formulae derived in Theorems 2 and 3 is a little inferior as compared to Theorem 1, that may be due to the numerical error attributed to the fact that only a

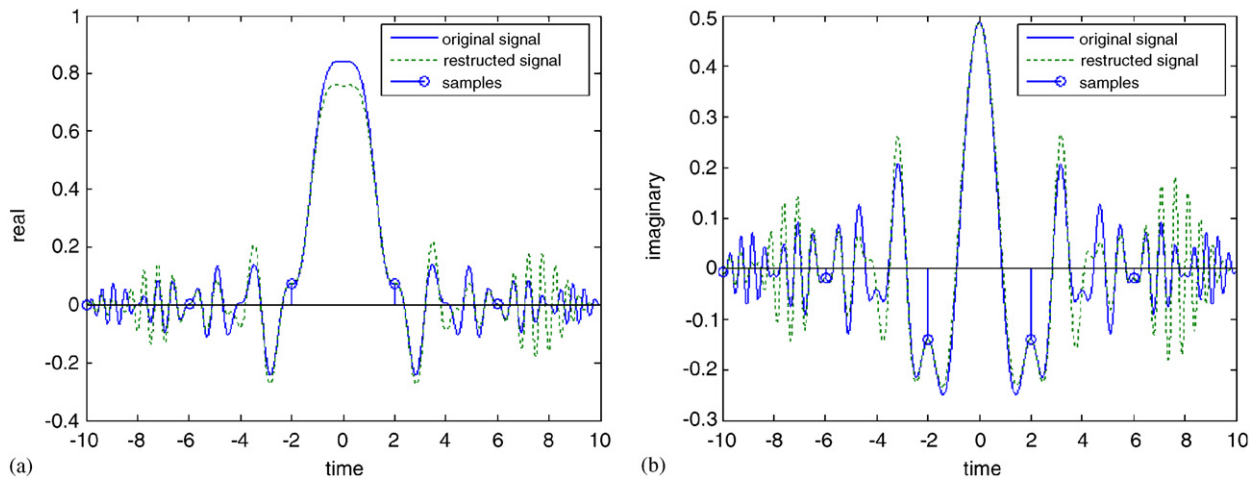


Fig. 3. (a) Real part of $f(t)$, of the samples of $f(t)$ and of the reconstructed signal by samples of the original signal and its first derivative at half the Nyquist rate of LCT domain. (b) Imaginary part of $f(t)$, of the samples of $f(t)$ and of the reconstructed signal by samples of the original signal and its first derivative at half the Nyquist rate of LCT domain. It is noted from Fig. 3 that the reconstructed signal denoted by dotted line overlaps almost exactly the original signal denoted by continuous line.

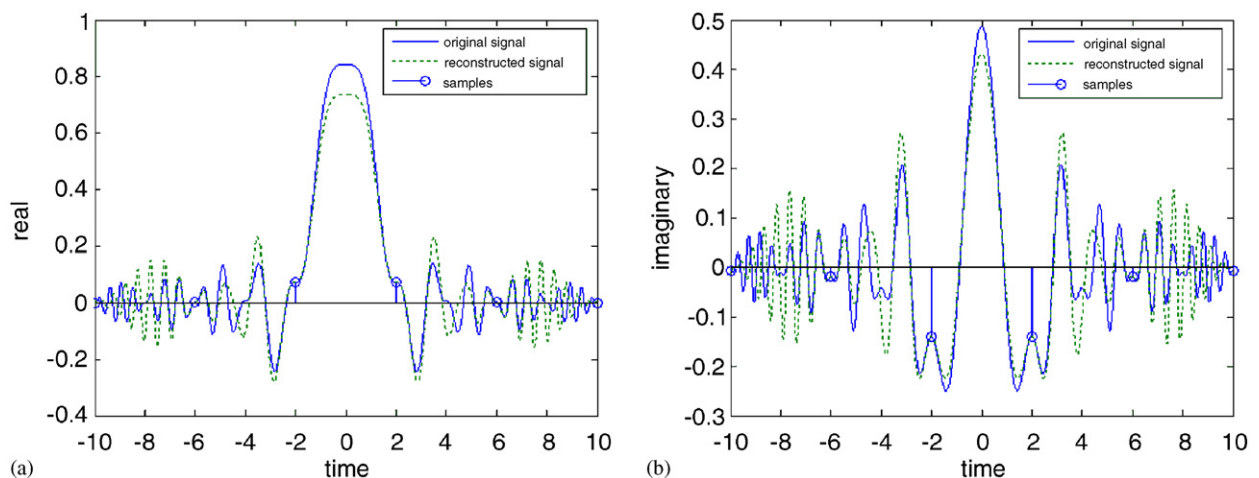


Fig. 4. (a) Real part of $f(t)$, of the samples of $f(t)$ and of the reconstructed signal by samples of the original signal and its generalized Hilbert transform at half of the Nyquist rate of LCT domain. (b) Imaginary part of $f(t)$, of the samples of $f(t)$ and of the reconstructed signal by samples of the original signal and its generalized Hilbert transform at half of the Nyquist rate of LCT domain. It is noted from Fig. 4 that the reconstructed signal denoted by dotted line overlaps almost exactly the original signal denoted by continuous line.

finite interval of the signal $f(t)$ and its derivate or GHT is processed during numerical calculation.

6. Conclusions and future works

The paper investigates the sampling theorem of band-limited signal associated with the linear canonical transform. Firstly, the sampling theorem representation of band-limited signals in linear canonical transform sense from the samples taken at Nyquist rate has been derived. Then, the other two signal representation formulae using samples taken at half the Nyquist rate from the signal and its first derivative or its generalized Hilbert transform in linear canonical transform are obtained. Finally, experimental results and discussions about these achieved results and its potential applications in linear canonical transform domain are proposed and obtained. The well-known results about the sampling theorem in Fourier transform domain, fractional Fourier transform domain can be looked as special cases of our achieved results. Future works along this path is to derive new interpolation formulae for non-uniformly sampled signals or high-dimensional signals and the applications of these results in real applications.

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